



Economics, School of Social Sciences
College of Humanities, Arts, & Social Sciences

HE2003 Econometrics I
AY2024/2025 Semester 1

Course Aims

This course aims to equip students with a basic understanding of econometric methods and models. It emphasizes the use of linear and nonlinear regression techniques in formulating and testing micro, macro, and financial economic hypotheses. Students will learn methods for estimating causal effects using real economic data, develop the ability to discern which econometric method is most appropriate in each situation and get some hands-on experience with data analysis using statistical software.

Intended Learning Outcomes (ILO)

By the end of this course, you will be able to:

1. Determine the functions, appropriateness, and theoretical assumptions of econometric methods and models.
2. Use linear and nonlinear regression techniques in formulating and testing micro, macro, and financial economic hypotheses.
3. Identify and apply methods for estimating causal effects using real economic data and use statistical package to analyse real-word data.
4. Discern which econometric method is most appropriate in a given situation.

Course Content

1. Simple Linear Regression Model, Ordinary Least Squares (OLS) Estimator, Least Square Assumptions
2. Properties of the OLS Estimator, Hypothesis Tests, and Confidence Intervals
3. Regression with Binary Variables, Heteroskedasticity
4. Omitted Variable Bias, Linear Regression with Multiple Regressors

5. Multicollinearity, Hypothesis Tests, and Confidence Intervals in Multiple Regression
6. Test of Joint Hypothesis, Chi-squared Distribution, F-statistic
7. Nonlinear Regression Functions: Polynomial and Logarithmic Models
8. Nonlinear Regression Functions: Interaction Models
9. Model Specification and Empirical/Data Issues

Teaching Faculty

	<u>Office</u>	<u>Telephone</u>	<u>E-mail Address</u>
Asst Prof WANG Wenjie	HSS-04-65	6316-8958	wang.wj@ntu.edu.sg
Office Hours:	By Appointment		

Prerequisite

MH1820 Introduction to Probability and Statistical Methods or same-level course

References and Reading

Main Reference: Lecture Slides and Notes

Supplementary Reference: James H. Stock and Mark W. Watson, *Introduction to Econometrics* (Third or Fourth Edition)

Statistical Software

STATA

Course Assessment

Grades on this course will be based on the following components.

Quiz: **20%**

Attendance and Participation: **10%**

Assignment (tutorial questions) and Presentation: **20%**

Final Examination: **50%**

TOTAL: 100%

Academic Integrity

Good academic work depends on honesty and ethical behavior. The quality of your work as a student relies on adhering to the principles of academic integrity and to the NTU Honour Code, a set of values shared by the whole university community. Truth, Trust, and Justice are at the core of NTU's shared values.

As a student, it is important that you recognize your responsibilities in understanding and applying the principles of academic integrity in all the work you do at NTU. Not knowing what is involved in maintaining academic integrity does not excuse academic dishonesty. You need to actively equip yourself with strategies to avoid all forms of academic dishonesty, including plagiarism, academic fraud, collusion, and cheating. If you are uncertain of the definitions of any of these terms, you should go to the [Academic Integrity website](#) for more information. Consult your instructor(s) if you need any clarification about the requirements of academic integrity in the course.

Planned Weekly Schedule

Week	Topic	Course LO	Readings/Activities
1	Introduction	1	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 1: 1.1, 1.2 Chapter 2: 2.1, 2.2, 2.3
2	Simple linear regression model 1	1,2	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 4: 4.1, 4.2
3	Simple linear regression model 2	1,2,3	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter

			4: 4.3, 4.4
4	Simple linear regression model 3	1,2,3	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 2: 2.5, Chapter 3: 3.1, Chapter 4: 4.3, 4.4
5	Simple linear regression model 4	1,2,3	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 3: 3.2, 3.3, Chapter 4: 4.5, Chapter 5: 5.3, 5.1
6	Simple linear regression model 5	1,2,3	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 3: 3.2, 3.3, Chapter 5: 5.1, 5.2, 5.4
7	Multiple linear regression model 1	1,2,3,4	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 6: 6.1, 6.2, 6.3
Recess Week			
8	Quiz		
9	Multiple linear regression model 2	1,2,3,4	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 6: 6.5, 6.6, 6.7
10	Multiple linear regression model 3	1,2,3,4	Main Reference: Lecture Slides/Notes

			Supplementary Reference: Stock and Watson, Chapter 7: 7.1, 7.2
11	Multiple linear regression model 4	1,2,3,4	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 7: 7.2, 7.3
12	Nonlinear regression model 1	1,2,3,4	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 8: 8.2
13	Nonlinear regression model 2 and Course Review	1,2,3,4	Main Reference: Lecture Slides/Notes Supplementary Reference: Stock and Watson, Chapter 8: 8.3 Tutorial questions and supplemental materials

*** The proposed schedule/content may be adjusted if it is in the best interest of student-learning.**